ELSEVIER

Earth and Planetary Science Letters 134 (1995) 203–217

EPSL

Paleomagnetic evidence for clockwise rotation of the Simao region since the Cretaceous: A consequence of India–Asia collision

Chen Haihong [a], Jon Dobson [b,1], Friedrich Heller [b], Hao Jie [a]

[a] *Institute of Geology, Chinese Academy of Sciences, Beijing 100029, China*
[b] *Institut für Geophysik, ETH-Honggerberg, CH-8092 Zurich, Switzerland*

Received 20 June 1994; accepted 23 June 1995

Abstract

Samples were collected for paleomagnetic analysis from 36 sites from six formations west of the Red River Fault in southwestern Yunnan Province, China on the Simao block (23.4°N, 100.8°E) in an attempt to examine the effects of the India–Asia collision on the Asian continent. Paleomagnetic results from rocks ranging from Cretaceous to Miocene in age indicate a large clockwise rotation of these sites with respect to stable Eurasia since the Late Cretaceous. The tectonic-corrected data from Cretaceous samples (mean direction of two formations) are $D = 112.6°$, $I = 34.7°$, $\alpha_{95} = 7.5°$, $k = 9.1$, $n = 45$, for Eocene samples the data are $D = 84.7°$, $I = 38.9°$, $\alpha_{95} = 7.6°$, $k = 12.0$, $n = 32$, and for Miocene samples they are $D = 21.1°$, $I = 35.5°$, $\alpha_{95} = 7.1°$, $k = 15.3$, $n = 29$. The samples from the Paleocene and Pliocene are either remagnetized or too weak to provide stable directions. The Cretaceous sites were sampled from folded beds and these sites pass the fold test. The amount of rotation seen in this study ($R = 86.1 \pm 9.9°$ as a mean value for the Cretaceous, $R = 76.8 \pm 11°$ for the Eocene, and $R = 14.7 \pm 10.0°$ for the Miocene) is significantly larger than that documented in previous studies from the Lower Cretaceous of northwestern and central Yunnan Province and may indicate that differential rotation has taken place in the region. This rotation can be explained by the northward motion and collision of India with Asia in the Paleocene, which caused continuous deformation of southwestern China in the form of strike-slip faulting and clockwise rotation. It is concluded that the rotation and deformation resulting from this collision probably continued from the Paleocene until at least the Miocene.

1. Introduction

The most important event of the Cenozoic in Asia is the collision of the Indian subcontinent with Eurasia 45 m.y. ago [1]. Geological investigations, together with laboratory experiments and theoretical analysis, indicate that this collision was probably responsible for crustal deformation of the Asian continent in the form of extensive strike-slip faulting as well as block rotations along the flanks of the collision zone and crustal shortening [2–9]. In the southern Tibetan plateau, large-scale crustal shortening created by northward indentation of the Indian plate since the end of the Cretaceous has been inferred from paleomagnetic data [10,11]. In addition, a narrow zone of 'domino-type' counterclockwise rota-

[1] Current address: Department of Physics, University of Western Australia, Nedlands, Perth, WA 6907, Australia

0012-821X/95/$09.50 © 1995 Elsevier Science B.V. All rights reserved
SSDI 0012-821X(95)00118-2

tions along the Yaluzangbu suture has been reported [7,8,12]. In the southeastern part of the plateau (i.e., the Three Rivers region), large-scale strike-slip faults with displacements of greater than 500 km have been recognized along the Ailaoshan–Red River left-lateral fault [3,13,14]. The timing of the left-lateral movements, which is constrained by isotope dating of granitic rocks from the fault, is placed between 35 and 19–17 Ma [13,14]. Large-angle clockwise rotations (28–65° with respect to stable Eurasia) have been revealed by paleomagnetic studies of Cretaceous rocks in the region and in nearby areas [15–19].

Despite all this study, the mode of crustal deformation of the continental lithosphere in the region remains controversial. Two models have been proposed: deformation of a rheological continuum [4,20,21] and a combination of crustal thickening and large-scale crustal extrusion [2,6]. The former puts forth the theory that the collision between India and Eurasia is absorbed by shortening and thickening with minor strike-slip faulting, while the latter stresses lateral crustal movements escaping along strike-slip faults. A key to evaluating the validity of the two models lies in the investigation of the scale of regional movements and the style and timing of crustal deformation, especially in the vicinity of large strike-slip faults. Paleomagnetism provides a powerful tool for investigating and quantifying these factors.

Previous paleomagnetic studies of Cretaceous rocks from two localities near the Red River Fault (the Yongping and Chuxiong basins, see Fig. 1)

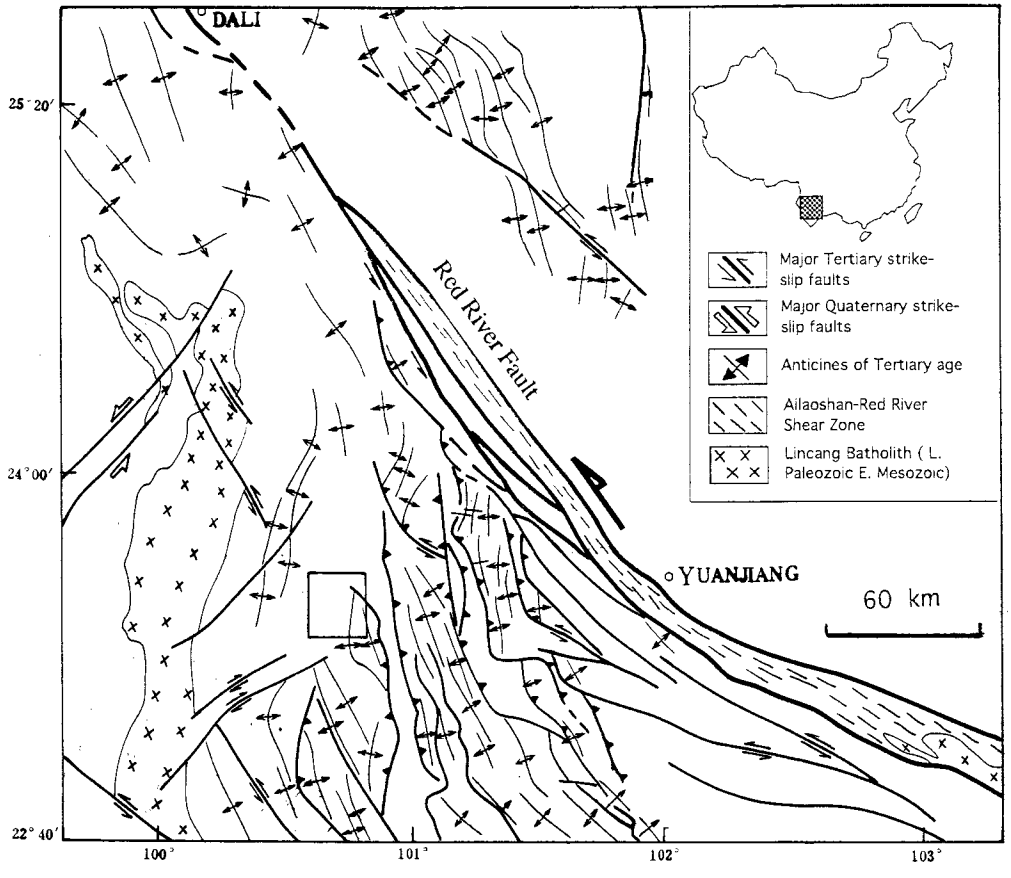


Fig. 1. (a) Structural map of the Simao block and neighbouring regions, showing major Cenozoic strike-slip faults, folds and fractures. The box is the sampling region shown in Fig. 2. (Simplified after [3].) The large arrow represents Tertiary movement along the Red River Fault. (b) Tectonic map of the Simao block and surrounding blocks and melanges.

reveal that the two areas have experienced a similar amount of clockwise rotation of 28 and 31° with respect to stable Eurasia [17,18] and 23–26° with respect to the eastern Yangtze [22,23]. These authors, therefore, reach the conclusion that a large area covering the two sampling regions along with the Red River Fault rotated as a single block, which could be explained by the model of rheological deformation of the continental crust. In contrast, larger differential rotations in different regions inferred using Jurassic and Cretaceous paleomagnetic data were reported in the Simao, Dali and Baoshan regions [15,16], which demonstrates a more complicated pattern of crustal disposition in Sundland. Moreover, late Mesozoic paleoreconstructions of northern Thailand suggest a southward translation of the region after the mid-Cretaceous, supporting the extrusion model [19]. In order to resolve this controversy surrounding the two models, more paleomagnetic studies, especially for the Cenozoic, are needed.

In an effort to determine the extent and temporal relationship of deformation caused by the India–Asia collision in the Three Rivers region, we have sampled rocks ranging in age from Late Cretaceous to

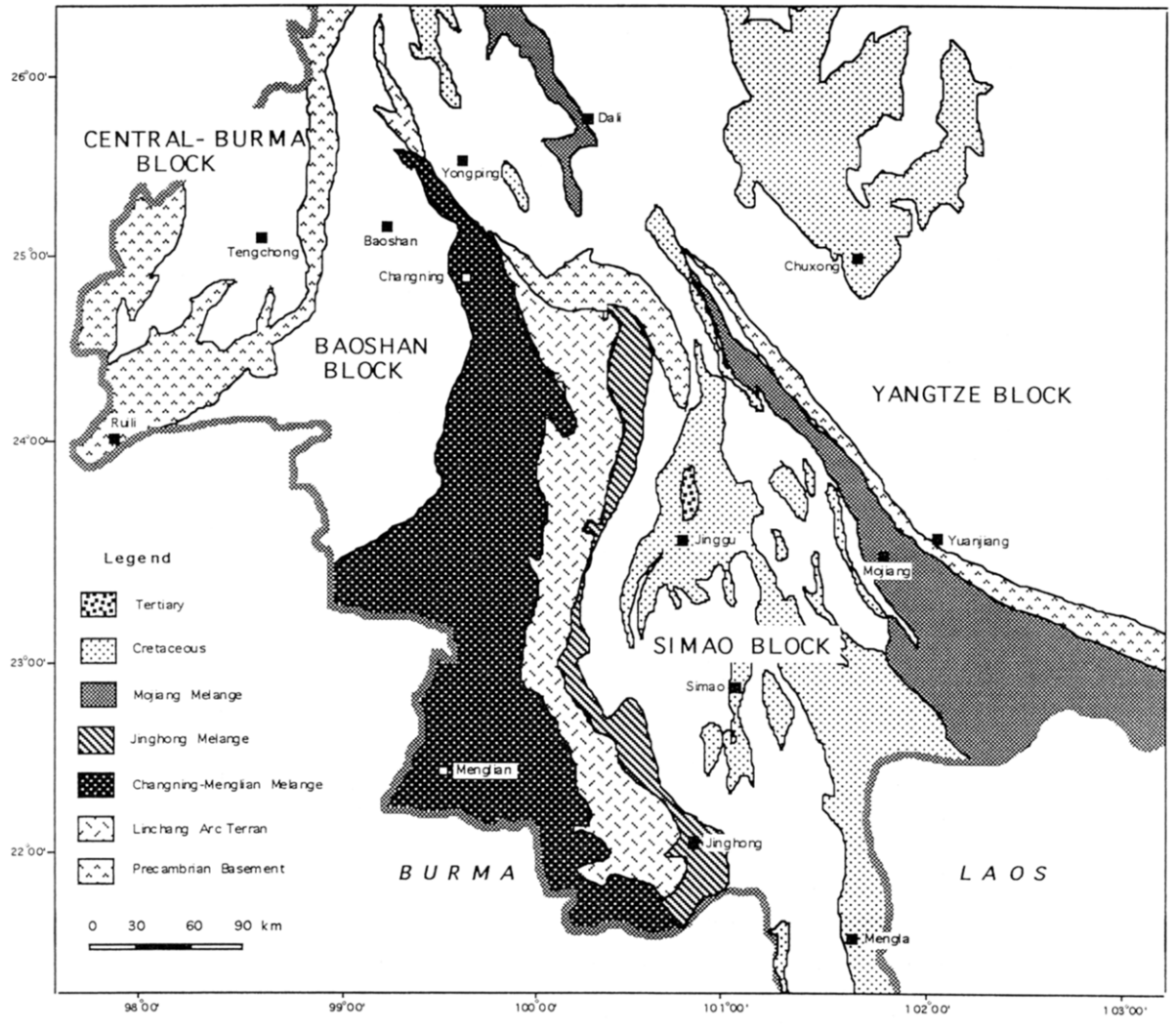


Fig. 1 (continued).

Miocene from the Simao area and of Pliocene age from the Baoshan area. A total of 232 samples were collected for paleomagnetic analysis from six formations.

2. Geology and sampling

The Three Rivers region is a tectonic collage of continental fragments and arcs from Gondwana and Cathaysia formed during the evolution of paleo-Tethys [24][25,26]. The blocks, such as the Baoshan block (or Shan–Thai–Malay), the Lincang arc, and the Simao (or Indochina) block, are bounded by melange zones. These zones are the Changning–Menglian Main paleo-Tethys Melange Zone between the Baoshan block and the Lincang arc, the Jinghong Melange Zone between the Lincang arc and the Simao block, and the Mojiang Melange Zone between the Simao and Yangtze blocks.

The Simao block is located in western Yunnan Province and is bounded in the northeast by the Red River Fault, which cuts and offsets the Mojiang suture zone and extends southwest to the Lincang

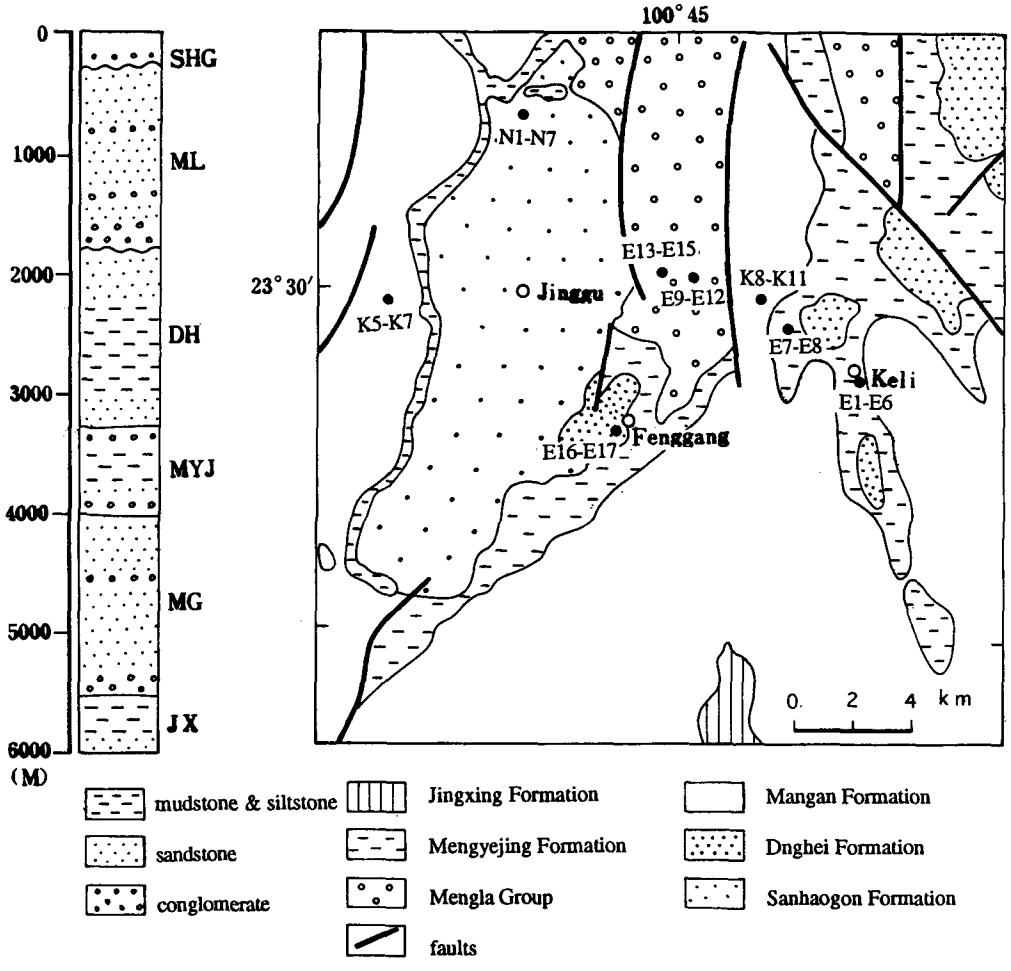


Fig. 2. Geological map of the Jinggu region. ● = Sampling locations. *SHG* = Sanhaogou Formation (Miocene); *ML* = Mengla Group (Eocene–Oligocene); *DH* = Denghei Formation (Eocene); *MYJ* = Mengyejing Formation (Paleocene); *MG* = Mangang Formation (Lower Cretaceous); *JX* = Jingxing Formation (Lower Cretaceous). The sampling location for the Jingxing Formation is off the map, at 23.4°N, 100.4°E.

batholith, a former arc terrane (Fig. 1). The region is extensively covered with Jurassic and Cretaceous continental redbeds with scattered Upper Paleozoic and Triassic marine deposits in the cores of anticlines. In Jinggu County (23.4°N, 100.8°E), relatively continuous Cenozoic strata are exposed in a N–S trending synclinorium, reaching a total thickness of about 4000 m [27].

The Lower Cretaceous Jingxing and Mangang Formations are composed of interbedded red sandstones and mudstone with conglomerates in the lowermost part. The two formations reach a total thickness of up to 1200 m in the region. Abundant freshwater fossils such as the bivalves *Kareanaia*, *Nippononaia*, *Plicatounio* and *Pergrinoconcha*, the ostracods *Cypriden*, *Jinggulla* and *Damonclla* and the gastropod *Estheria charophyte*, together with sporopollenin, indicate an Early to mid-Cretaceous age. Seven sites from the Mangang Formation (23.5°N, 100.8°E) (Fig. 2) and four sites from the Jingxing Formation (23.4°N, 100.4°E), including sites for a fold test, were sampled with a portable water-cooled drill.

The Paleocene Mengyejing Formation unconformably overlies the Lower Cretaceous sediments and is dominated by redbeds with salts and gypsum in the upper part. Charophyte fossils such as *obtusochara*, *tectochara* and *rhabdochara* are found in the strata, implying an early to middle Paleocene age. Eight sites from this formation were taken from road cuttings (23.5°N, 100.8°E) northeast of the town of Jinggu.

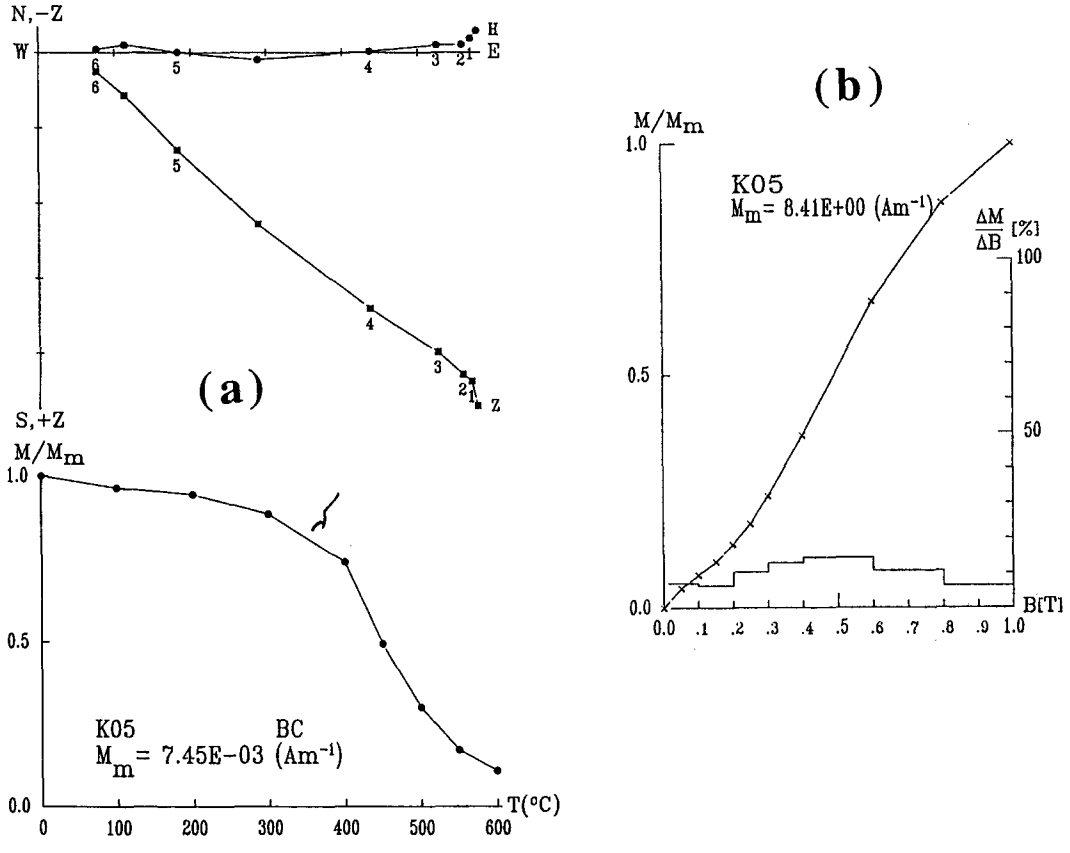


Fig. 3. (a) Zijderveld diagram for a single-component sample from the Lower Cretaceous Jingxing Formation. Numbers next to the Zijderveld projections represent temperature ×100°C. Z = Vertical projection, H = horizontal. M_m = NRM magnetization. (b) IRM acquisition curve for the Jingxing Formation. M_m = magnetization at 1 T. B[T] = Field in Tesla.

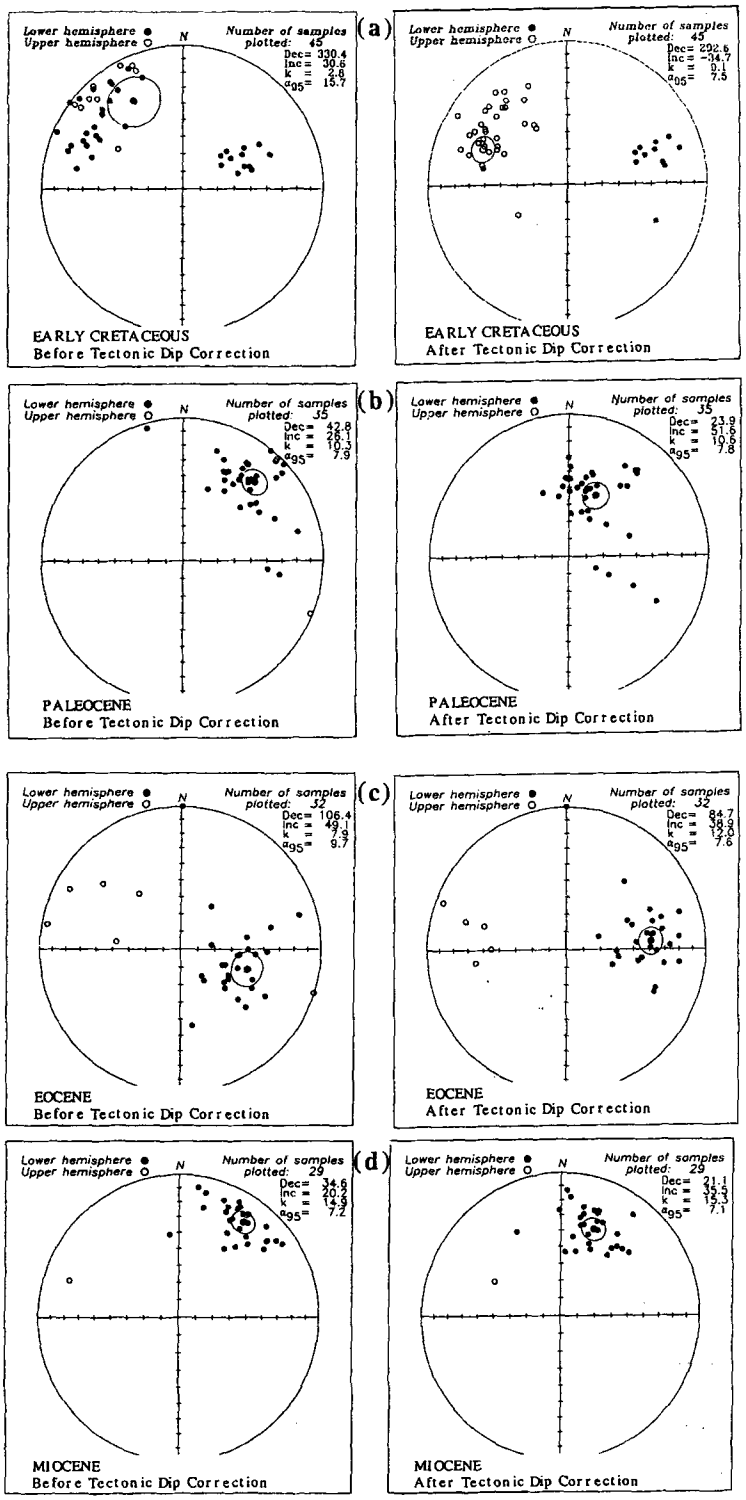

(a)
Lower hemisphere ●
Upper hemisphere ○
N
Number of samples plotted: 45
Dec= 330.4
Inc= 30.6
k = 2.6
α95= 15.7
EARLY CRETACEOUS
Before Tectonic Dip Correction
Lower hemisphere ●
Upper hemisphere ○
N
Number of samples plotted: 45
Dec= 292.6
Inc= -14.7
k = 0.1
α95= 7.5
EARLY CRETACEOUS
After Tectonic Dip Correction
(b)
Lower hemisphere ●
Upper hemisphere ○
N
Number of samples plotted: 35
Dec= 42.8
Inc= 26.1
k = 10.3
α95= 7.9
PALEOCENE
Before Tectonic Dip Correction
Lower hemisphere ●
Upper hemisphere ○
N
Number of samples plotted: 35
Dec= 23.9
Inc= 51.6
k = 10.6
α95= 7.8
PALEOCENE
After Tectonic Dip Correction
(c)
Lower hemisphere ●
Upper hemisphere ○
N
Number of samples plotted: 32
Dec= 106.4
Inc= 49.1
k = 7.9
α95= 9.7
EOCENE
Before Tectonic Dip Correction
Lower hemisphere ●
Upper hemisphere ○
N
Number of samples plotted: 32
Dec= 84.7
Inc= 38.9
k = 12.0
α95= 7.6
EOCENE
After Tectonic Dip Correction
(d)
Lower hemisphere ●
Upper hemisphere ○
N
Number of samples plotted: 29
Dec= 34.6
Inc= 20.7
k = 14.9
α95= 7.2
MIOCENE
Before Tectonic Dip Correction
Lower hemisphere ●
Upper hemisphere ○
N
Number of samples plotted: 29
Dec= 21.1
Inc= 45.5
k = 15.3
α95= 7.1
MIOCENE
After Tectonic Dip Correction

Table 1
Paleomagnetic results for the Lower Cretaceous Jingxing and Mangang Formations, the Paleocene Mengyujing Formation, the Eocene–Oligocene Mengla Group and the Miocene Sanhaogou Formation from Jinggu

Location					In Situ	Tilt Corrected	VGPs
Lat (N)/Lon (E)	Formation	Bedding	n/N	Site	Dec/Inc/k/α95	Dec/Inc/k/α95	Lat(N)/Lon(E)
23.4/100.4	Jingxing	97/19	10/19	3	78.4/58.0/49.4/17.7	84.4/39.6/49.1/17.8	13.6/171.5
23.4/100.5	Mangang	265/64	13/16	3	294.2/23.1/299.5/7.1	297.5/-33.3/301.7/7.1	
23.5/100.8	Magnang	350/46	17/20	4	309.3/-6.2/58.9/12.1	294.4/-38.1/58.8/12.1	
	Mangang mean		40/55	7	303.1/6.4/18.0/14.6	295.8/-36.0/92.2/6.3	-13.9/161.3
23.5/100.8	Mengyejing	262/36	35/57	9	42.8/26.1/10.3/7.9	23.9/51.6/10.6/7.8	48.5/185.1*
23.5/100.7	Mengla	50/23	32/40	7	106.4/49.1/7.9/9.7	84.7/38.9/12.0/7.6	13.2/172.2
23.5/100.7	Sanhaogou	267/29	29/48	6	34.6/20.2/14.9/7.2	21.1/35.5/15.3/7.5	70.0/183.2

n/N is the number of samples used for statistics/all samples collected. *Site* is the number of sites sampled. *Dec* and *Inc* are the declination and inclination; k is the Fisher precision parameter; α_{95} is the radius of the cone of 95% confidence. * Pole position calculated using the in-situ direction.

The Eocene Denghei Formation is a fine-grained sequence dominated by mudstone and siltstone. Its age is estimated by stratigraphic correlations with the fossiliferous beds in the neighbouring regions because no fossils have been found in the study area. This formation appears to be seriously weathered, and only twenty samples were taken from a natural outcrop at the Fenggang Salt Mine.

The Eocene to Oligocene Mengla Group is also a redbed sequence and again is unfossiliferous. Its age is estimated by stratigraphic contacts with the fossiliferous strata of the underlying Paleocene Mengyejing Formation and the overlying Miocene Sanhaogou Formation. Seven sites were sampled from this group (23.5°N, 100.7°E).

The Miocene Sanhaogou Formation, which gently dips to the northwest, consists of yellow to grey sandstones interbedded with grey siltstone and laminated mudstone. The age is determined by the presence of the well-known Jinggu flora, which dates from the latest Oligocene to Early Miocene [28]. Seven sites from this formation were sampled from a road cutting (23.5°N, 100.7°E) located 5 km north of Jinggu.

Twelve samples were taken from Pliocene sediments at an outcrop located 40 km southeast of Baoshan (24.8°N, 99.5°E). This formation is not named on the geological maps but consists of flat-lying, semi-consolidated, grey lacustrine clays and siltstone with laminations.

All samples were subjected to stepwise thermal demagnetization with a TSD-1 thermal specimen demagnetizer and the directions were analyzed using standard paleomagnetic techniques [29–31]. Measurements were made with a 2G cryogenic magnetometer shielded with Helmholz coils interfaced with a Hewlett–Packard HP-1000 computer. In addition, acquisition of isothermal remanent magnetization (IRM) was measured in fields up to 1 T with an Oxford Instruments electromagnet in order to determine the dominant coercivities of the magnetic minerals in the samples.

3. Results

3.1. Lower Cretaceous

Stable, high-temperature directions were isolated during thermal demagnetization in 40 out of 56 samples from the Jingxing and Mangang Formations. The remainder of the samples exhibited unstable

Fig. 4. Equal-area projections of geographic and bedding-corrected characteristic directions from (a) the Cretaceous Jingxing and Mangang Formations, (b) the Paleocene Mengyejing and Yunlong Formations, (c) the Eocene to Oligocene Mengla Group, and (d) the Miocene Sanhaogou Formation

behaviour during thermal treatment. NRM intensities of the stable samples are generally between 10^{-3} and 10^{-2} A/m. IRM acquisition and thermal demagnetization of representative samples reveals the dominance of a high-coercivity, high-unblocking-temperature phase (i.e., hematite) (Fig. 3). Thermal demagnetization of the NRM shows that blocking temperatures are around 600–650°C. Demagnetization vectors usually became unstable at temperatures above 600°C. In some samples, a single component appears in the Zijderveld diagrams after the removal of a low-temperature component by 200°C. Directions of the low-temperature component are unstable, and are probably a mixture of the recent field and either a laboratory VRM or the primary magnetization. Directions of the high-temperature component (300–650°C) are stable and represent the characteristic remanent magnetisation (ChRM), with easterly downward magnetizations.

Samples from the Mangang Formation exhibited a demagnetization behaviour that was similar to that of the Jingxing Foramtion, although the overprint persists to higher temperatures. In these samples the high-temperature component defined between 500 and 650°C is the ChRM. The site-mean ChRM directions are plotted for both formations in equal-area projection diagrams (Fig. 4). The ChRM directions of the Mangang Formation pass a fold test at the 99% confidence level [32]. The normal directions of the Jingxing are not, however, antipodal to the reversed ones of the Mangang Formation (see Table 1). The reasons for this will be discussed in the next section. The tilt-corrected formation mean direction for the Jinggu Formation from this locality is $D/I =$ 84.4/39.6° with $\alpha_{95} = 17.8$, corresponding to a paleomagnetic pole at 13.6°N, 171.5°E. The formation mean direction for the Mangang Formation is D/I $= 295.8/-36.0°$ with $\alpha_{95} = 6.3$, yielding a paleomagnetic pole at 13.9°S, 161.3°E.

3.2. Paleocene

The samples from the Paleocene Mengyejing Formation were taken from a thick, massive, well-sorted and porous fine-grained red sandstone sequence of possible aeolian origin. The majority of the samples exhibit more or less univectorial behaviour during thermal treatment (Fig. 5), only a few displaying a resolvable low-temperature component, which is eliminated below 350°C. The in-situ ChRM direction of these samples is northeasterly with a gentle inclination (Fig. 4), similar to those of the Miocene (shown later). The tilt-corrected direction is northerly but steeper than the present-day Earth field (PDF). The mean direction of this formation before tilt correction is $D/I = 42.8/26.1°$ with $\alpha_{95} = 7.9°$, yielding a paleomagnetic pole at 48.5°N, 196.4°E; this most probably represents a remagnetization during the Miocene or slightly earlier.

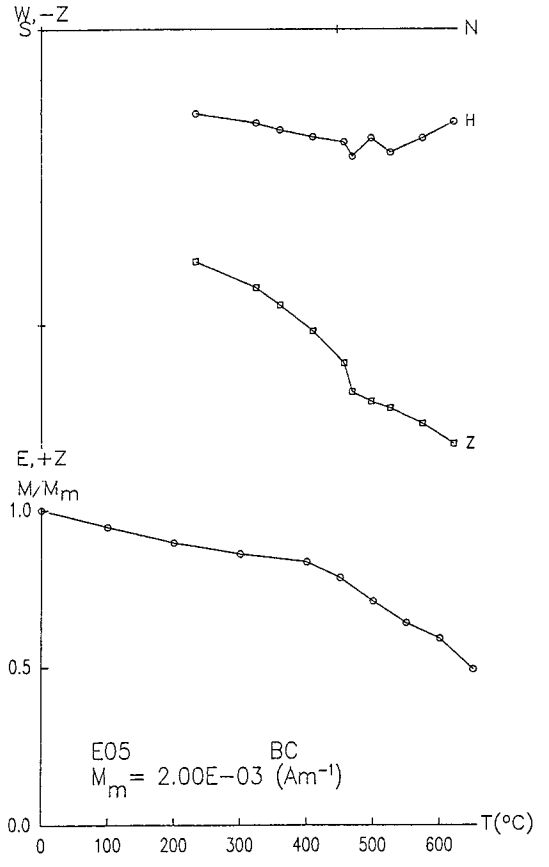


Fig. 5. Zijderveld diagram for the Paleocene Mengyejing Formation. Z = Vertical projection, H = horizontal. M_m = NRM magnetization.

3.3. Eocene

All samples from the Denghei formation were either too weak in NRM intensity to provide stable

high-temperature directions or remagnetized in the PDF. This behaviour was not totally unexpected because the outcrop appeared weathered. No significant paleomagnetic results were isolated from these samples.

3.4. Eocene–Oligocene

The samples from the Eocene–Oligocene Mengla Group preserve a stable remanent magnetization. NRM intensities of these samples range from 10^{-2} to 10^{-3} A/m. Hematite is the major carrier of magnetization, as indicated by the high coercivity and the high blocking temperature spectrum from the IRM measurements (Fig. 6b). Two, and sometimes three, components can be recognized in most samples (Fig. 6a): a low-temperature component (below 200°C) is probably a VRM acquired in the recent field, a component defined between 300 and 500°C is similar to the Miocene ChRM direction, probably representing a remagnetization in this period, and a high-temperature component (500–675°C) includes antipodally normal and reversed polarities (Fig. 4). The grouping of the directions improves after correcting for bedding tilt, yielding $D/I = 84.7/38.9°$ with $\alpha_{95} = 7.6°$, which corresponds to a paleomagnetic pole at 13.2°N, 209.2°E.

3.5. Miocene

Of the 42 Miocene samples, stable components of magnetization were revealed in 29 and unblocking temperatures just above 500°C were observed (Fig. 7a). The remaining samples exhibited unstable behaviour during thermal demagnetization. IRM acquisition curves indicate a dominance of low-coercivity

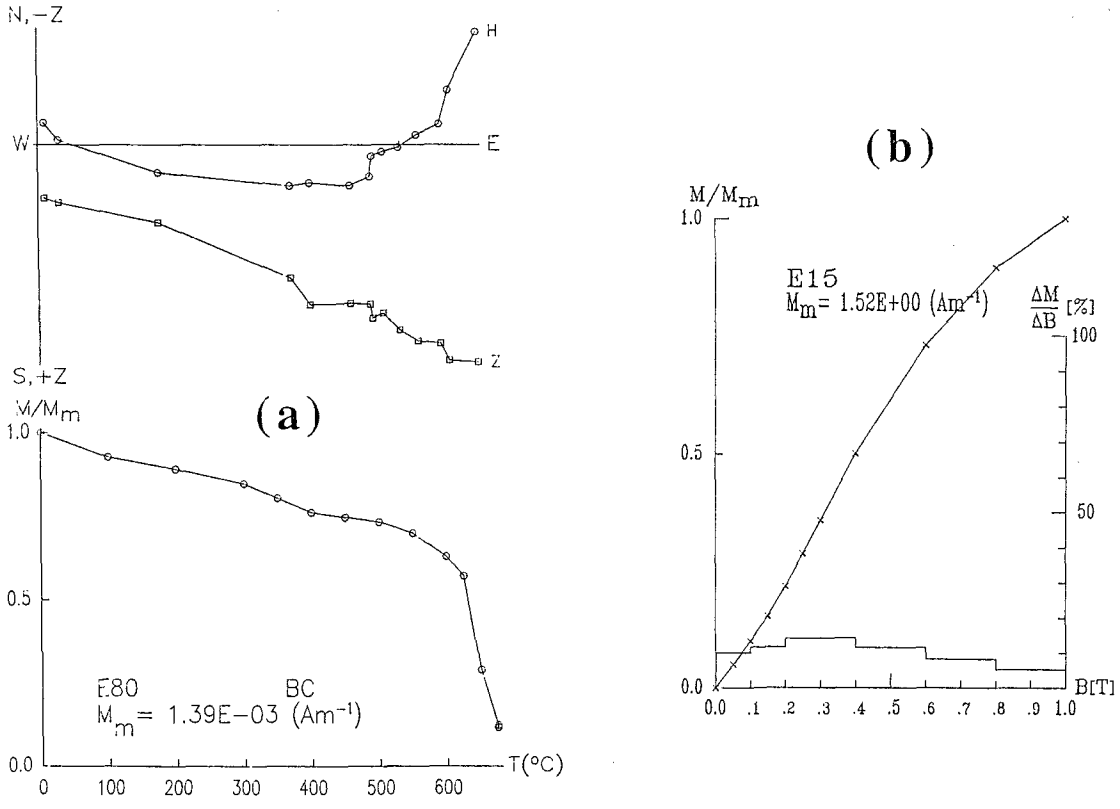


Fig. 6. (a) Zijderveld diagram for the Eocene to Oligocene Mengla Group. Z = Vertical projection, H = horizontal. M_m = NRM magnetization. (b) IRM acquisition curve from the same group. M_m = magnetization at 1 T. B[T] = Field in Tesla.

minerals (Fig. 7b). This information, along with the lower unblocking temperatures, indicates that the remanent magnetization is probably carried by magnetite. NRM intensities of the stable samples range from 10^{-4} to 10^{-3} A/m. Remanent directions below 300°C are clearly affected by the PDF, while directions between 300 and 450°C move towards the origin. Further thermal demagnetization creates either new mineral phases, causing an increase in remanent intensities, or unstable directions due to low remaining intensities. These northeasterly downward directions were resolved after removal of the PDF overprint and are probably of primary origin (Fig. 4), with a mean tilt-corrected direction of $D/I = 21.1/35.5°$ with $\alpha_{95} = 7.1°$, yielding a paleomagnetic pole at 70.0°N, 197.8°E.

3.6. Pliocene

Only ten pilot samples were used for demagnetization. All samples carried a PDF overprint with unblocking temperatures of about 150°C, indicating that the remanence is carried by goethite. Above these temperatures, no stable directions could be isolated.

4. Discussion

4.1. Stability of magnetization

The ChRM isolated from the Lower Cretaceous Mangang Formation passes the fold test and is thus pre-folding (pre-Eocene) in age, although the two sampling localities are not from the same fold and are about 15 km apart. However, the reversely magnetized direction from this formation is in contrast with those of the Lower Cretaceous Jingxing Formation, which all exhibit normal magnetization and are not antipodal to the Mangang directions. The reliability of the Jingxing result is not as certain because fold and reversal tests are lacking. Nevertheless, this

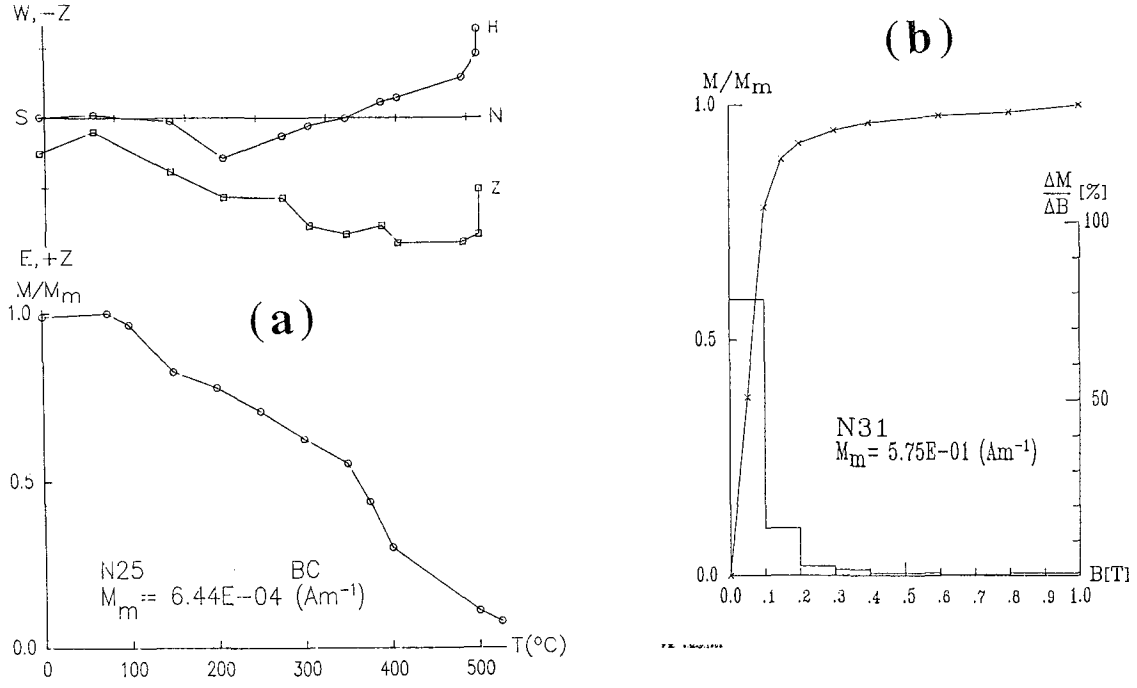


Fig. 7. (a) Zijderveld diagram from the Miocene Sanhaogou Formation. Z = Vertical projection, H = horizontal. M_m = NRM magnetization. (b) IRM acquisition curve from the same formation. M_m = Magnetization at 1 T. B[T] = Field in Tesla.

result from the Jingxing Formation is consistent with those reported earlier for the Mangang Formation from a nearby area [16]. A discrepancy of nearly 35° in the declination in our Mangang results and the declinations of the Jingxing Formation and the results from [16] exists. There are two possible interpretations for this problem: Eocene to Oligocene remagnetization of the rocks from the Jingxing and Mangang in [16], or differential rotations by local thrust faults between sampling localities. It is difficult to determine which interpretation is correct, but local rotation is possible because several notable thrust faults demarcate the sampling locations (see Fig. 1).

The ChRM direction in geographic coordinates of the Paleocene Mengyejing Formation is similar to the corrected ChRM results of the Miocene samples, although a slight easterly deviation is seen. The texture of the samples collected from this formation is unique, being porous, well sorted, and massive in structure, which implies an aeolian origin in a lacustrine environment. Migrating thermal fluids can easily penetrate such porous rock bodies, resulting in thorough remagnetization. An important tectonic event occurred in the late Oligocene in the region, resulting in deformation and uplift of the earlier sequence. Miocene sediments, which occur as limited basin fills, unconformably overlie the folded basement. It seems reasonable to consider that the Paleocene samples have been remagnetized as a result of this tectonic event.

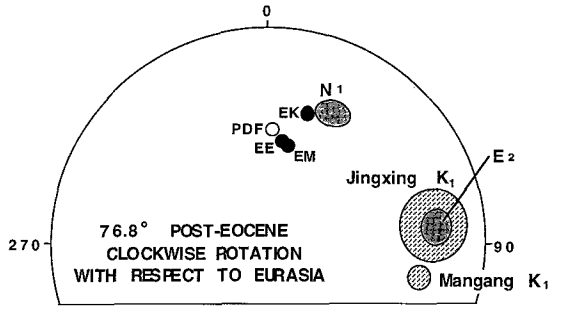


Fig. 8. Stereographic projection of the formation means with α_{95} confidence limits. K_1, E_2 and N_1 are Lower Cretaceous, Eocene–Oligocene and Miocene respectively. *EK*, *EE* and *EM* are expected directions of Lower Cretaceous (*EK*), Eocene–Oligocene (*EE*) and Miocene (*EM*) calculated from the VGPs of 90, 40 and 20 Ma for Eurasia [33]. PDF is the present-day Earth field direction for the Jinggu area.

The ChRM from the Eocene–Oligocene Mengla Formation is quite different from the PDF and Miocene directions, with two antipodal polarities. This seems to be an indication of primary magnetization. This direction is similar to those of the Lower Cretaceous Jingxing Formation from this study and the Mangang Formation in [16].

The reliability of the ChRM of the Miocene Sanhaogou Formation is not as certain because we lack fold and reversal tests. However, the thermal demagnetization revealed that a northeasterly direction per-

Table 2
Amount of rotation in the Jinggu area and neighbouring regions since the Cretaceous with respect to stable Eurasia

Location (Lat[N]/Long[E])	Age	Eurasia Pole (Ma)	D(obs)	D(exp)	R	Reference
Jinggu (23.5/ 100.7)	Miocene	20	24.1 ± 8.7	6.4 ± 4.9	14.7 ± 10.0	This Study
Jinggu (23.5/100.7)	Eocene-Oligocene	40	84.7 ± 9.8	7.9 ± 5.3	76.8 ± 11.1	This Study
Jinggu (23.4/100.4) (Jingxing)	Lower Cretaceous	90	84.4 ± 17.8	14.0 ± 4.0	70.4 ± 23.3	This Study
Jinggu (23.5/100.7) (Mangang)	Lower Cretaceous	90	115.8 ± 6.3	14.0 ± 4.0	101.8 ± 7.8	This Study
Jinggu (23.4/100.9)	Middle Cretaceous	80	79.4 ± 12.5	14.6 ± 3.6	64.8 ± 13.0	16
Mengla	Middle Cretaceous	80	60.8 ± 9.6	14.5 ± 3.6	46.3 ± 10.3	16
Dali (25.6/100.2)	Middle Cretaceous	70	6.9 ± 12.5	14.1 ± 4.5	-7.2 ± 13.6	16
Chuxiong (25.0/101.5)	Lower Cretaceous	90	44.6 ± 14.2	17.1 ± 5.4	27.5 ± 15.2	17
Yongping (25.5/99.5)	Lower Cretaceous	90	42.0 ± 25.2	13.9 ± 4.2	27.9 ± 25.8	18

D(obs) and D(exp) are the observed and expected declination. The expected value is calculated from the Eurasian APWP [33]. R is the amount of clockwise rotation with respect to stable Eurasia (negative values represent counterclockwise rotation) [$R = D_{obs} - D_{exp}$]. Estimate of uncertainties follows [18].

sists after removal of heavy overprints of the PDF. This direction, which is not consistent with the present overprint at the sampling location, is probably primary.

4.2. Amount of rotation since the Cretaceous

Large clockwise deflections in declination with respect to stable Eurasia appear in the results for the Lower Cretaceous and Eocene–Oligocene samples (Fig. 8). These deviations probably result from a process of clockwise rotation of the region. Table 2 lists all the values of rotation calculated from this study and the surrounding regions. The amount of rotation R is determined by $R = D_{obs} - D_{exp}$, where D_{obs} and D_{exp} are the observed and expected declination of the Jinggu area. The expected declination value is calculated from the VGPs of 90, 40 and 20 Ma for Eurasia [33]. The uncertainty ΔD is given by $\Delta D = \sin^{-1}(\sin \alpha_{95}/\cos I)$, while the uncertainty in rotation ΔR is given by $\Delta R = [(\Delta D_{obs})^2 + (\Delta D_{exp})^2]^{1/2}$, where ΔD_{obs} and ΔD_{exp} are uncertainties at the 95% confidence level [18].

The amount of rotation calculated from the Lower Cretaceous Jingxing results is 70.4° with respect to stable Eurasia, and 101.8° from the Mangang data in this study (Table 2). As we discussed earlier, the difference in rotation magnitude between the two formations may be produced by local rotations or Eocene–Oligocene remagnetization of Jingxing. A mean rotation value for these two formations is 86.1° but it is likely that the Mangang data are representative of the block because the fold test constrains the magnetization to the pre-Eocene. The amount of rotation seen in the Eocene–Oligocene Mengla samples is 76.8° with respect to stable Eurasia, similar to the value documented in the Jingxing area of this study and the Mangang results of [16]. Clockwise rotation of 14.7° with respect to stable Eurasia is obtained from the Miocene Sanhaogou Formation.

4.3. Tectonic implications

Whereas it is difficult to determine which Cretaceous pole from this study is more representative of the block, both the Jingxing and Mangang results are consistent with significant rotation. The closeness of these Cretaceous results to those for the Eocene–Oligocene may indicate that most significant rotation took place after Eocene–Oligocene time. A large displacement is observed between these poles and the eastern Yangtze pole (72°N, 199°E, $\alpha_{95} = 7°$) [18] and the 90 Ma pole for Eurasia (77°N, 200°E, $\alpha_{95} = 3°$) [33], clearly suggesting clockwise rotation of the Jinggu area since the Cretaceous.

These large-scale clockwise rotations seen in the Cenozoic in the Simao region are likely to be related to the continental collision of the Indian plate with Eurasia and the subsequent orogeny. Paleomagnetic investigations in the last decade have shown that southern Tibet in the northern part of the area has been shifted northward from the Cretaceous to the Pliocene, with latitudinal changes from 6°S to 47°N [10–12,34]. The continental lithosphere of the southern margin of the Eurasian continent was extensively imbricated and thickened due to the continuous indentation of the Indian continent. In the eastern part of the Tibetan plateau (the Three Rivers region), bending, strike-slip faulting and rotation of the continental crust are observed without extensive crustal thickening. This process has been tested by a simple extrusion model experiment [2]. This effect may reach as far as the opening of the South China Sea.

The Three Rivers region is a tectonic collage of continental fragments and arcs from Gondwana and Cathaysia that formed during the paleo-Tethys stage [24–26]. These blocks are bounded by several melange zones, such as the Changning–Menglian Main paleo-Tethys Melange Zone, the Jinghong Melange Zone and the Mojiang Melange Zone [35–37]. The presence of easily mobilized melange zones between the tectonic blocks in this region seems to have been favourable for the initiation of strike-slip faulting and block rotations when the indentation of the Indian plate began.

The large clockwise rotation of the region was recognized during previous paleomagnetic investigations [16–18]. The maximum observed rotation is about 35° greater than that predicted [16,38] with a more complicated rotation pattern. Huang and Opdyke [16] listed all the Cretaceous paleomagnetic data available from Tibet and the surrounding area and presented a thorough discussion of their tectonic implications. We agree with the conclusions that they proposed—that Sundaland was not deformed as a coherent block in response to the collision of India

with Asia and the subsequent northward movement of the Indian continent. It should be noted, however, that many strike-slip faults, both major and minor, are extensively developed in this region, with significant differences in the magnitude of clockwise rotations since Cretaceous times (28° with respect to stable Eurasia in the Yongping basin [18], 46.3° in the Mengla area, 64.8° in the Jinggu area [16], 76.8° for the Eocene–Oligocene data in the Jinggu basin (this study), 14.4° in northern Thailand [16], 31° in the Chuxiong basin [17], and 7.2° of counterclockwise rotation in Dali [16]). All of these areas appear to be bounded by strike-slip faults. Along the Red River Fault, for example, a large difference in rotation magnitude of about 40° between the Chuxiong and Simao regions is observed. Even within small areas, such as between Yongping (on the western side of the Red River Fault) and Dali (on the eastern side), rotation magnitude may vary due to the contribution of local strike-slip faulting (see Fig. 1).

Aside from the clockwise rotation of the region, lateral extrusion of continental blocks along large-scale strike-slip faults is also revealed by some paleomagnetic data. About 880 ± 580 km of post-mid-Cretaceous translation, for example, was suggested by the Cretaceous paleomagnetic data from the Khorat basin of northern Thailand [19], while about 220–480 km for south Yunnan was estimated based on the Cretaceous results from Dali, Jinggu and Mengla [16]. In the latter case the calculated translations are within the paleomagnetic uncertainties.

The expected paleolatitude computed for the Cretaceous to Miocene at Jinggu is 20.6° (calculated from the VGP of Eurasia at 90 Ma) and 28.8° for Eurasia at 20 Ma. The observed paleolatitudes derived from the Cretaceous to Miocene results of this study are lower than the expected values from the VGPs of Eurasia and previous studies [16]. The observed paleolatitude from the Miocene data is 19.6°, 9.6° less than the expected value. This discrepancy is probably a result of lateral extrusion of the study area.

Regarding the temporal relationship of the rotation, our data suggest that active movements associated with clockwise rotations and strike-slip faulting can be as young as Miocene. This is consistent with the estimate of 17 Ma for the latest activity of left-lateral strike-slip faulting dated from deformed inclusions within the Ailaoshan–Red River shear zone [13,14].

5. Conclusions

(1) The Jinggu area has been rotated clockwise, apparently by more than 90°, with respect to stable Eurasia since the Early Cretaceous. Discrepancies in rotation magnitudes are seen in different sampling locations, implying local rotations or remagnetization.

(2) Since the Eocene–Oligocene, the Jingu area has been rotated clockwise through 76.8° with respect to Eurasia.

(3) A clockwise rotation of 14.7° is revealed from the Miocene samples, indicating that the rotation processes were active at least until the Miocene.

(4) The available Cretaceous paleomagnetic data from the Jinggu area and neighbouring regions demonstrate the presence of differential rotations in different sampling localities, usually bounded by strike-slip faults. This implies that the Three Rivers region was not deformed as a coherent block in response to the collision of India with Asia and the subsequent northward indentation.

Acknowledgements

We would like to thank Prof. William Lowrie, Profs. Sun Shu, Li Jiliang and Cong Belin and Prof. K.J. Hsü for their help with this project. We are grateful to Beat Geyer for his assistance in the laboratory. The original manuscript had been significantly improved by the comments of two anonymous reviewers. This project is supported by Swiss National Funds grant 21-29834.90 and the Chinese Academy of Sciences. [*UC*]

References

[1] B.D. Johnson, C.M. Powell and J.J. Veevers, Spreading history of the eastern Indian Ocean and Giant India's northward flight from Antarctica and Australia, Geol. Soc. Am. Bull. 87, 1560–1566, 1976.

[2] P. Tapponnier, G. Peltzer, A. Le Dain, R. Armijo and P. Cobbold, Propagating extrusion tectonics in Asia: new in-

sights from simple experiments with plasticine, Geology 10, 611–616, 1982.
[3] P. Tapponnier et al., The Ailao Shan/Red River metamorphic belt: Tertiary left-lateral shear between Indochina and South China, Nature 343, 431–437, 1990.
[4] P. Tapponnier and P. Molnar, Slip-line field theory and large-scale continental tectonics, Nature 264, 320–324, 1976.
[5] G. Houseman and P. England, Finite strain calculations of continental deformation 1. Methods and general results for convergent zones, J. Geophys. Res. 93, 15085–15117, 1986.
[6] G. Peltzer and P. Tapponnier, Formation and evolution of strike-slip faults, rifts, and basins during the India–Asia collision: an experimental approach, J. Geophys. Res. 93, 15085–15117, 1988.
[7] Y. Otofuji, S. Funahara, J. Matsuo, F. Murata, T. Nishiyama, X. Zheng and K. Yaskawa, Paleomagnetic study of western Tibet: deformation of a narrow zone along the Indus Zhangbo suture between India and Asia, Earth Planet. Sci. Lett. 92, 307–316, 1989.
[8] Y. Otofuji, J. Kadoi, S. Funahara, F. Murata and X. Zheng, Paleomagnetic study of the Eocene Quxu pluton of the Gangdese belt: crustal deformation along the Indus–Zangbo suture zone in southern Tibet, Earth Planet. Sci. Lett. 107, 369–379, 1991.
[9] P. Molnar and P. Tapponnier, Cenozoic tectonics of Asia: Effects of a continental collision, Science 189, 419–426, 1975.
[10] J. Besse, V. Courtillot, J. Pozzi, M. Westphal and Y. Zhou, Paleomagnetic estimates of crustal shortening in the Himalaya thrusts and Zangbo suture, Nature 311, 621–626, 1984.
[11] J. Achache, V. Courtillot and Y. Zhou, Paleogeographic and tectonic evolution of southern Tibet since middle Cretaceous time: new paleomagnetic data and synthesis, J. Geophys. Res. 89, 10311–10399, 1984.
[12] Y. Otofuji, Y. Inoue, S. Funahara, F. Murata and X. Zheng, Paleomagnetic study of eastern Tibet—deformation of the Three Rivers region, Geophys. J. Int. 103, 85–94, 1990.
[13] T. Harrison, W. Chen, P. Leloup, F. Ryerson and P. Tapponnier, An early Miocene transition in deformation regime on the Red River fault zone, Yunnan and its implication for Indo-Asian tectonics, J. Geophys. Res. 97, 7159–7182, 1992.
[14] P. Leloup, T. Harrison, F. Ryerson, W. Chen, Q. Li, P. Tapponnier and R. Lacassin, Structural, petrological and thermal evolution of a Tertiary ductile strike-slip shear zone, Diancangshan, Yunnan, J. Geophys. Res. 98, 6715–6743, 1993.
[15] K. Huang and N. Opdyke, Paleomagnetic results from the Upper Carboniferous of the Shan–Thai–Malay block of western Yunnan, China, Tectonophysics 192, 333–344, 1991.
[16] K. Huang and N. Opdyke, Paleomagnetic results from Cretaceous and Jurassic rocks from Southwest Yunnan: evidence for large clockwise rotations in the Indochina and Shan–Thai–Malay terranes, Earth Planet. Sci. Lett. 117, 507–524, 1993.
[17] S. Funahara, N. Nishiwaki, M. Miki, F. Murata, Y. Otofuji and Y. Wang, Paleomagnetic study of Cretaceous rocks from the Yangtze block, central Yunnan, China: implication for the India–Asia collision, Earth Planet. Sci. Lett. 113, 77–91, 1992.
[18] S. Funahara, N. Nishiwaki, F. Murata, Y. Otofuji and Y. Wang, Clockwise rotation of the Red River fault inferred from paleomagnetic study of Cretaceous rocks in the Shan–Thai–Malay block of western Yunnan, China, Earth Planet. Sci. Lett. 117, 29–42, 1993.
[19] Z. Yang and J. Besse, Paleomagnetic study of Permian and Mesozoic sedimentary rocks from Northern Thailand supports the extrusion model for Indochina, Earth Planet. Sci. Lett. 117, 525–552, 1993.
[20] P. England and D. McKenzie, A thin viscous sheet model for continental deformation, J. Geophys. Res. 70, 295–321, 1982.
[21] S. Cohen and R. Morgan, intraplate deformation due to continental collisions: a numerical study of deformation in a thin viscous sheet, Tectonophysics 132, 247–259, 1976.
[22] D. Kent, G. Xu, K. Huang, W. Zhang and N. Opdyke, Paleomagnetism of upper Cretaceous rocks from South China, Earth Planet. Sci. Lett. 79, 179–184, 1986.
[23] R. Enkin, Z. Yang, Y. Chen and V. Courtillot, Paleomagnetic constraints on the geodynamic history of China from the Permian to the Present, J. Geophys. Res. 97, 13953–13984, 1992.
[24] J.Q. Huang and B.W. Chen, The Evolution of the Tethys in China and Adjacent Regions, 109 pp., Geol. Publ. Ho., Beijing, 1987,
[25] D. Zhong, Paleotethys: Lithospheric Tectonic Evolution of Western Yunnan and Sichuan, in press, Science Press, Beijing, 1994 (in Chinese, with English Summ.).
[26] K.J. Hsu, Geology of Switzerland and an Introduction to Tectonic Facies, Princeton University Press, 1995.
[27] Bureau of Geology and Mineral Resources of Yunnan Province, Geological map of the Jinggu area (1:200,000), 1981.
[28] Bureau of Geology and Mineral Resources of Yunnan Province, Regional Geology of Yunnan Province (with 1:500,000 Geological Map), 727 pp., Geol. Publ. Ho., Beijing, 1990.
[29] R. Fisher, Dispersion on a sphere, Proc. R. Soc. London A217, 295–305, 1953.
[30] J.A.D. Zijderveld, AC demagnetization of rocks: analysis of results, in: Methods in Paleomagnetism, D.W. Collinson et al., eds., pp. 254–286, Elsevier, Amsterdam, 1967.
[31] J. Kirschvink, The least squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc. 62, 699–718, 1980.
[32] M.W. McElhinny, Statistical significance of the fold test in paleomagnetism, Geophys. J. R. Astron. Soc. 8. 338–340, 1964.
[33] J. Besse and V. Courtillot, Revised and synthetic apparent polar wonder paths of the African, Eurasian, North American and Indian Plates, and true polar wonder since 200 Ma, J. Geophys. Res. 96, 4029–4050, 1991.
[34] C. Klootwijk, P. Conaghan and C. Powell, The Himalayan arc: large scale subduction, oroclinal bending and back-arc spreading, Earth Planet. Sci. Lett. 75, 167–183, 1985.

[35] X. Duan and H. Zhao, On the Ailaoshan–Tengtiaohe fracture —a subduction zone for ancient plates, Acta Geol. Sin. 55, 258–266, 1981 (in Chinese, with English Abstr.).

[36] J. Li, Tectonic evolution of the Sanjiang belt, western Yunnan, Sci. Geol. Sin. 4, 337–346, 1988 (in Chinese, with English Abstr.).

[37] B. Cong, G. Wu, Q. Zhang, R. Zhang, M. Zhai, D. Zhao and W. Zhang, Petrotectonic evolution of the Paleotethys belt in western Yunnan, China, Sci. China (B) 23(11), 1202–1207, 1993 (in Chinese).

[38] P. Tapponnier, G. Peltzer and R. Armijo, On the mechanics of the collision between India and Asia, in: Collisional Tectonics, M.P. Coward and A.C. Ries, eds., Geol. Soc. London Spec. Publ. 19, 115–157, 1986.